

NAME

Maitreya Gupta



Personal

DOB: 02.10.2000
Hobbies: Karate, KickBoxing,
Game Dev, Coding
Languages: Hindi, English,
Bengali

Technical Skills

Programming:
Python, C++, Java, C
Web:
React, Flask, Django,
HTML/CSS
Robotics:
ROS, ESP32, MediaPipe, YOLO
Tools:
Git, VS Code, Godot, Three.js

Social Links

- GitHub
- HackerRank
- TryHackMe
- LinkedIn
- Portfolio Website

Achievements

- State-level Bronze Winner in KickBoxing
- Gold Winner in Ju Srijan (Tech Fest of Jadavpur University)

OBJECTIVE

M.Tech student in Intelligent Automation & Robotics with strong expertise in Computer Vision, Robotics, Full-Stack Development, and Interactive Simulations. Seeking AI/ML, Robotics, and Software Engineering roles where I can apply my research and development skills.

EDUCATION

2023–Present	M.Tech (Intelligent Automation and Robotics) , Jadavpur University CGPA: 8.67/10, YGP: 82.75%
2019–2023	B.Tech (Electronics and Communication) , St. Thomas College CGPA: 8.33/10
2017–2019	Patha Bhavan (12th: 65%)
2015–2017	Patha Bhavan (10th: 83.28%)

SELECTED CERTIFICATIONS

- Fullstack Development - Coding Ninjas
- Python (Basic) - HackerRank
- SQL (Basic) - HackerRank
- BSNL Telecom Training

TECHNICAL ASSESSMENTS

- TCS iON National Qualifier Test (NQT) 2026**
 - Cognitive Ability: 82.85% | Advanced Quantitative & Reasoning: 93.97%
 - Programming (Python): 72.76%
 - Official Scorecard

PROJECTS

Full-Stack & Mobile Development

- Kanban Connection Board** — MERN Stack, React, Node.js, Express.js, MongoDB, JWT, Hugging Face API, Three.js
Live Demo **GitHub** **Admin Portal** **Admin GitHub** *(Backend available on request)*
 - Developed a full-stack Kanban productivity platform featuring secure user authentication, task management, and interconnected task visualization via a Connection Board.
 - Implemented AI-powered chatbot assistance and intelligent task suggestions leveraging the Hugging Face API.
 - Added drag-and-drop task management for intuitive movement of tasks across workflow stages.
 - Built secure JWT-based authentication with persistent task storage and automatic task creation timestamps.
 - Developed an administrative dashboard for monitoring user activity, reviewing tasks, tracking creation times, and flagging inappropriate content.
 - Enhanced user experience with responsive UI designs and Three.js visual elements.
 - Designed collaborative board functionalities to enable multi-user teamwork.
- Android Job Posting Platform** — Java, Kotlin, MERN Stack **Showcase**
 - Built a full-stack job platform with secure REST APIs using Node.js, Express, and MongoDB.
 - Integrated CSV bulk upload functionality for efficient job posting management.
 - Developed a responsive Android UI with seamless client-server integration.

Game Development & Simulation

- **DexterGallery** — React, Three.js, WebGL **Live Demo** **GitHub**
 - Built an immersive 3D interactive simulation platform for exploring diverse scientific domains, including Physics, Chemistry, Astronomy, Robotics, and Quantum Mechanics.
 - Developed a Discovery Hub with real-time 3D lab simulations, enabling hands-on virtual experimentation across multiple disciplines.
 - Designed modular lab environments (Radiation Physics, Optical Deflection, Semiconductor, Electromagnetism, Kinematics, Mathematics, and Computer Science) for educational and research purposes.
 - Designed real time 3D IoT labs interactive with the environment using OpenCV Python under IOTHub.
 - Implemented a React-based frontend with seamless navigation and responsive UI for an intuitive user experience.
- **Industrial Multi-Agent Robot Navigation Simulator** — Unity 6, C# **Demo Video**
 - Developed a server-authoritative multiplayer simulator featuring A*, Dijkstra, RRT, and NavMesh navigation algorithms.
 - Integrated support for Python-trained PyTorch models to execute custom navigation behaviors.
 - Implemented full multiplayer synchronization utilizing Netcode for GameObjects.
- **Inception Battlegrounds** — Roblox Studio, Lua **Play**
 - Designed a multiplayer combat experience with custom gameplay systems, progression mechanics, and UI.
 - Executed full lifecycle development from initial design to final deployment.

Robotics & AI

- **CRMSF: Vision-Based Gesture-Controlled Robotic Arm** — Python, TensorFlow, MediaPipe, YOLO, ROS, Gazebo **Demo**
 - Developed a real-time vision-based robotic teleoperation system for intuitive control of a 6-DOF robotic arm and dexterous gripper using monocular hand tracking.
 - Designed the **CNN Regressive Motion Stabilization Filter (CRMSF)** to reduce hand-tracking jitter and improve the stability and smoothness of robotic motion.
 - Integrated MediaPipe hand tracking, YOLO-based object perception, and ThreeJS WebGL simulation into an end-to-end gesture-to-actuation pipeline for low-latency robotic control.
 - Research manuscript currently under preparation.
- **MobiUNet-3D: Hybrid Brain Tumour Segmentation and Volumetric Reconstruction** — TensorFlow, Keras, Python **Live Demo**
 - Developed a lightweight hybrid deep learning framework with a shared MobileNetV2 encoder for simultaneous brain tumour classification and MRI segmentation, reducing redundant computation through multi-task learning.
 - Designed a unified segmentation-classification architecture with dedicated task-specific heads, enabling efficient feature reuse while maintaining a compact model footprint.
 - Implemented a volumetric reconstruction pipeline that converts segmented MRI slices into patient-specific STL models using the Marching Cubes algorithm for three-dimensional visualisation.
 - Integrated preprocessing, data augmentation, and K-fold cross-validation for robust model evaluation. Research manuscript under review.
- **Robotics Car Project** **Drive**
 - Implemented IR sensor-based obstacle detection with real-time data streaming to backend.
 - Created a Godot Engine interface with Python for data visualization and processing.

BACHELORS FINAL YEAR PROJECT

• Compact Wideband MIMO Antenna for 5G and IoT

Report

- Designed and simulated a compact four-element octagonal microstrip MIMO antenna using **CST Studio Suite**.
- Achieved **2.64–20 GHz** bandwidth, **>20 dB isolation**, **6.53 dB peak gain**, and **ECC < 0.01**.
- Fabricated and validated prototype in an anechoic chamber.

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